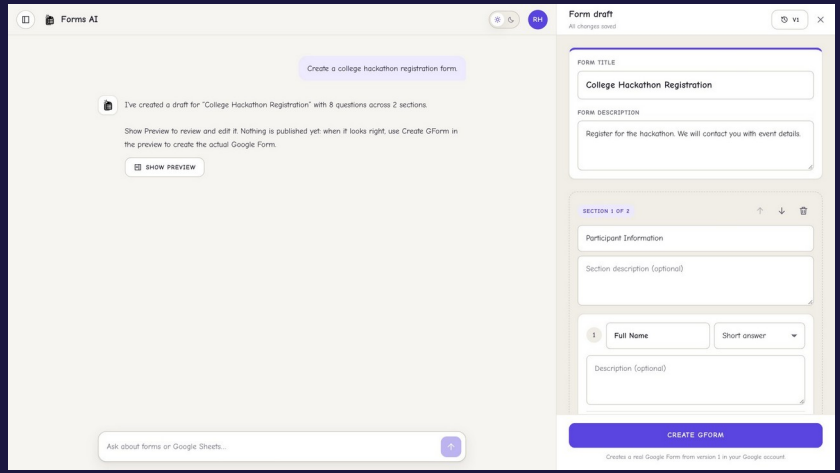
Forms AI

From one sentence to a live Google Form, and Google Sheets you can talk to.



github.com/rahulhansrajde/forms-ai

The problem: forms and sheets eat up time

Who has it

- Teachers and college clubs
- Hackathon and event organisers
- HR teams, NGOs, small businesses
- Mostly non-technical people

Why it matters

- Forms are built click by click
- Missed or badly typed questions mean messy data
- Sheets are fixed cell by cell, with errors
- AI chat gives text, not a real form

Four steps from idea to published form



01

Describe

"Create a college hackathon registration form."



02

Review

An editable draft opens beside the chat



03

Refine

"Remove phone number." Each change is a new version

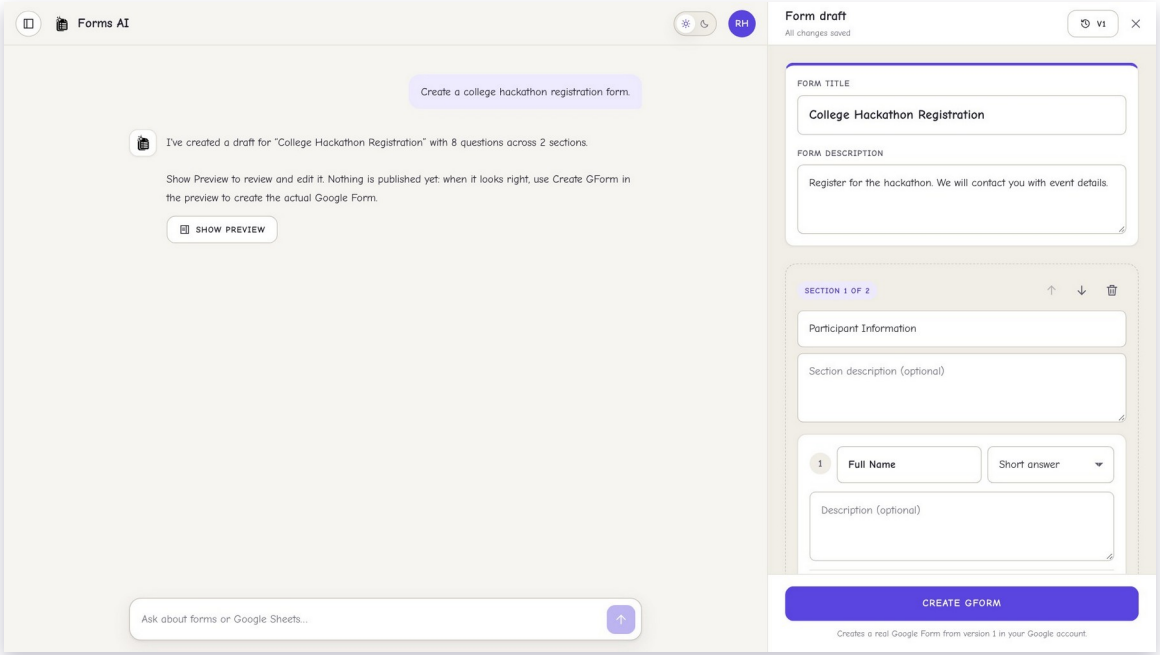


04

Publish

Create GForm: live link, editor link and QR code

Every request becomes an editable draft



Edit anything

Titles, sections, question types, choices, Required

Sections

Long forms are split into logical sections

Versions

Every edit is saved; switch with the version picker

Refine it by just asking

Remove phone number.

Removed "Phone Number" · version 2

What else can I ask?

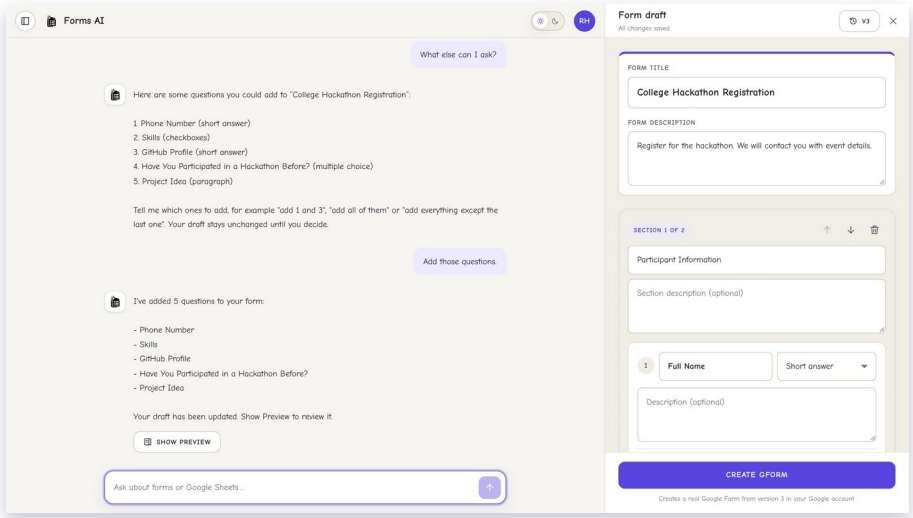
5 suggested questions; draft left unchanged

Add those questions.

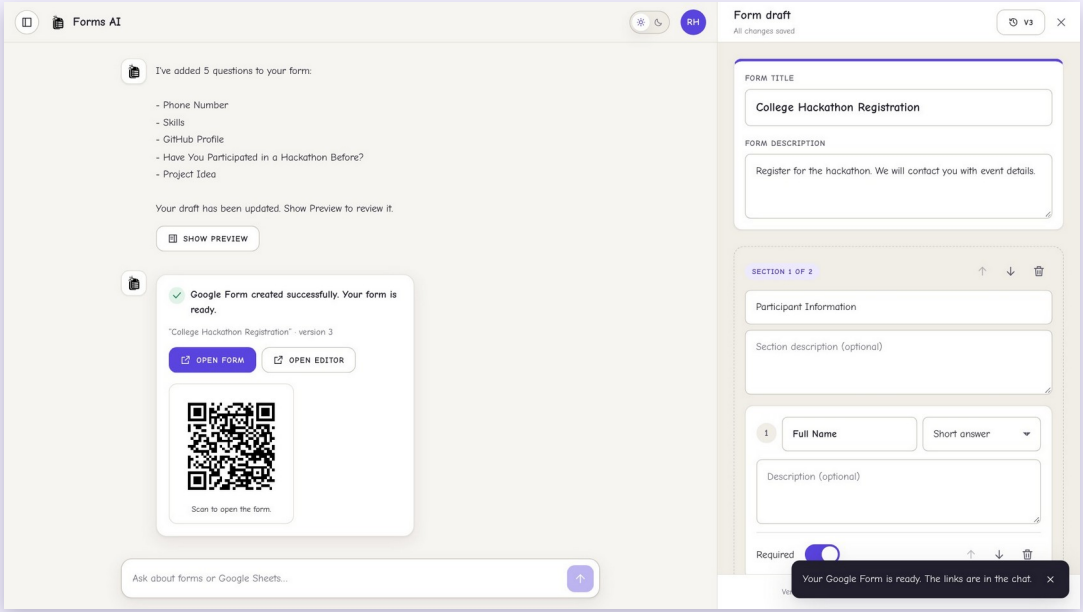
Added 5 questions · version 3

Start over with a shorter version.

Regenerated; old version kept in history



Publishing is one confirmed click



Connect once
Google OAuth for Forms and Sheets

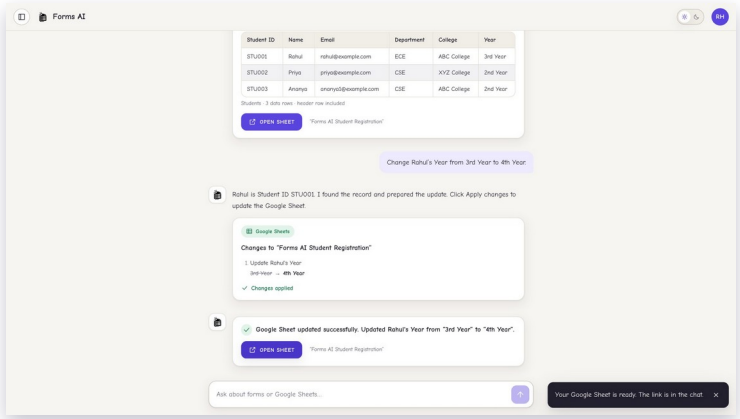
Confirm
Dialog shows the account and version

Share
Open form, Open editor, QR code

Retry safely
A failed publish keeps the draft

Google Sheets you can talk to

You type	Forms AI does
Use this Google Sheet: <URL>	Finds it, shows tabs; you click Connect
Show me the first 3 rows	Reads instantly into a table
What is Priya's department?	Answers a single-record question
Change Rahul's Year to 4th Year	Finds STU001, prepares the change; you Apply
Change Ananya's Year…	Two matches, so it asks for the Student ID



Reads happen immediately. Writes happen only after you click Apply changes.

Safe by design

Human in the loop

The AI never publishes or writes on its own. Forms and Sheets change only through buttons; destructive edits ask twice.

Encrypted tokens

Google OAuth tokens are encrypted at rest with AES-256-GCM. Passwords are hashed with scrypt.

Hardened API

HTTP-only session cookies, origin guard, Helmet headers and CSP, per-route rate limits.

Least surprise

Ambiguous records prompt for an ID instead of guessing. Failed publishes keep the draft.

A clean, layered TypeScript monorepo

Frontend
React 19 · Vite 7 · React Router

Shared contracts
Form schema · operations · intents

Tests
Vitest · Testing Library · Playwright

Backend · Node.js

Presentation
Express 5 routes, controllers, validators

Application
Use cases: chat, drafts, publish, sheets

Domain
Entities, value objects, interfaces

Infrastructure
Postgres/PGlite · AI · Google APIs · security



PostgreSQL 17
PGlite for dev & tests



Amazon Bedrock
or offline rule-based AI



Google APIs
OAuth · Forms · Sheets

Tested end to end

567

unit and integration tests passing

15

Playwright browser journeys

320–1440

px widths checked for overflow

2

themes: light and dark

npm run verify runs typecheck, lint, all tests and the production build: the same gate CI uses before every deploy.

Push to main, and it deploys to Azure



Push
main branch or manual run

Validate
verify + Postgres 17 tests + Playwright

Package
Production build zipped as one release

Deploy
OIDC sign-in → Azure App Service

Key App Service settings

`NODE_ENV=production · DATABASE_URL + DATABASE_SSL=true · TOKEN_ENCRYPTION_KEY`
`FRONTEND_ORIGIN + FRONTEND_DIST_DIRECTORY · GOOGLE_CLIENT_ID / SECRET / REDIRECT_URI`
`TRUST_PROXY=true · optional BEDROCK_API_KEY · startup command: npm start`

Run it in a minute

No Google or AI keys needed: embedded database and rule-based AI.

```
git clone https://github.com/rahulhansrajde/forms-ai && cd forms-ai
npm install
echo "DATABASE_URL=pglite://.data/forms-ai" > .env
echo "TOKEN_ENCRYPTION_KEY=$(openssl rand -base64 32)" >> .env
npm run dev        # open http://localhost:3000
```

Full setup, usage and deployment guide: README.md